Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years of professional experience building high-traffic, scalable web applications. Proven expertise in modern web engineering using React, Next.js (SSR, SSG, RSC), TypeScript, and GraphQL backend-for-frontend (BFF) architectures. Adept at optimizing rendering and loading times for performance at scale, implementing automated testing strategies (Playwright, Jest), and continuously improving CI/CD pipelines. Strong background in cross-functional collaboration, architectural ownership, and mentoring developers to drive complex features from design to production.

CORE COMPETENCIES

- **Frontend Architecture:** React, Next.js (SSR, SSG, RSC), TypeScript, component-based systems, micro-frontends.
- **Backend & APIs:** GraphQL, NestJS, NodeJS REST APIs, Python, schema evolution, authentication, caching.
- **Performance Optimization:** Code splitting, lazy loading, Lighthouse, React Profiler, bundle analysis.
- **Testing & CI/CD:** Playwright, Jest, Cypress, GitHub Actions, Docker, automated testing strategies.
- **Cloud & Infrastructure:** AWS, Vercel, Traefik, Docker Compose, PostgreSQL, MongoDB.
- **Leadership & Collaboration:** Cross-functional squads, architectural decisions, code reviews, mentoring.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a Vue.js web application to a **React** Chrome extension, optimizing workflows for clinical site integrations and enhancing adoption.
- Co-developed a full-stack **Next.js** chat widget powered by a RAG system using OpenAI GPT models, LangChain, and Vercel AI SDK, enabling users to query their data in real time via a conversational interface.
- Integrated internal and external APIs using **GraphQL** and REST, managing authentication, caching, and schema evolution to ensure seamless data flow between the frontend and backend services.
- Optimized application performance by implementing lazy loading, code splitting, dynamic imports, and CDN asset delivery, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Implemented JWT-based authentication and role-based access control in **Next.js** with **TypeScript** and React Query, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Increased release reliability by integrating **Playwright** end-to-end tests into the **CI/CD pipeline**, automating reporting with Allure Reports on GitHub Pages and configuring Slack notifications.
- Architected a reusable design system in React and Radix UI with Tailwind CSS, improving UI consistency, accessibility (WCAG), and developer velocity.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through rigorous code reviews.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a 25% boost in user engagement and collaborating with architects to create a scalable frontend architecture that accelerated development by 30%.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by 30-40% through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a 35% faster load time.
- Drove a 40% improvement in application performance by optimizing caching mechanisms across the React application.
- Established engineering standards using ESLint, Prettier, Husky, Lint-staged, and **GitHub Actions**, reducing errors and continuously improving **CI/CD pipelines**.
- Enhanced development workflow by introducing automated testing frameworks including **Jest** and Enzyme, leading to a 20% decrease in bug reports.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria

Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume bank Settlement Solution in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by 80%.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, Redux, RxJS, and Axios.
- Built a real-time account reconciliation solution, boosting daily reconciliation by 60% through innovative transaction processing strategies.

- Adopted the native EventSource API for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS** RESTful APIs and translated business requirements into efficient MongoDB database models and schemas.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to Heroku and AWS.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an Nx monorepo managing 6 applications: **NestJS** API backend, **Next.js** citizen-facing web app (22 chart types with Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Built the **Next.js** citizen-facing frontend with React 19, Tailwind CSS v4, and Recharts, featuring dark mode, URL state management, and responsive mobile layouts.
- Configured **CI/CD** with **GitHub Actions** for automated builds and Docker Compose production deployment with multi-stage Dockerfiles, health checks, memory limits, and 30s graceful shutdown.
- Implemented SSE streaming with 15s keepalive heartbeats, PostgreSQL 16 with Prisma v7, phone OTP authentication, and shareable public conversations with SEO optimization (SSR).
- Integrated **Playwright** E2E tests across 4 test projects (web auth, dashboard auth, desktop Chrome, mobile Pixel 5).

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*